

PSYCH-BENCH: A Comprehensive Cognitive Battery for Evaluating Holographic Liquid Brain Architectures

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Abstract

We present PSYCH-BENCH, a psychological benchmark suite comprising 88+ benchmarks across 26 cognitive domains for evaluating Holographic Liquid Brain (HLB) architectures—systems combining Hyperdimensional Computing, Liquid Time-Constant networks, Integrated Information Theory, and Active Inference. Existing AI evaluation frameworks lack psychometric infrastructure (HELM), cover narrow domains (ARC), omit normative human comparison (BIG-Bench), or ignore neuromodulatory effects (CogBench). PSYCH-BENCH addresses all four gaps with: (1) implementations of established paradigms (Stroop, WCST, N-back, and 85+ others) with psychometric assessment infrastructure including ICC reliability, speed-accuracy tradeoff curves, and adaptive staircases; (2) normative anchoring against 130+ published human baselines enabling z-score and clinical-band interpretation; (3) a 17-benchmark neuromodulator evaluation framework testing dose-response curves, tolerance/withdrawal dynamics, behavioral knockouts, and multi-transmitter synergy; and (4) a 14-indicator consciousness evaluation derived from six theories. All stimuli share a unified holographic encoding via five HDC adapter types, enabling cross-modal comparison. Demonstration on the SYMTHAEA HLB reveals a grand mean z-score of +1.095 across 26 cognitive domains (26/26 above human mean, 0 below), with cross-seed stability $SD = 0.019$ (3 seeds). Particular strengths emerge in Clinical ($z = +2.00$), InstitutionalReasoning ($z = +2.25$), SustainedAttention ($z = +2.19$), Speech ($z = +2.13$), Motor ($z = +2.12$), and Security ($z = +2.00$). Psychometric analysis reveals a gradient of reliability from Excellent to Poor (6 Excellent, 2 Good, 7 Moderate, 1 Poor across 16 evaluated benchmarks), selective

sensitivity to working-memory and encoding-noise ablation, and accuracy-pressure tradeoffs
in 4 of 8 tested benchmarks. Implemented in $\sim 20,000$ lines of Rust with 1,119+ unit tests,
PSYCH-BENCH is open-source and fully deterministic.

Keywords: cognitive benchmark, holographic computing, liquid neural networks, neuromodulation,
psychometrics, consciousness evaluation, hyperdimensional computing

1 Introduction

A new class of cognitive architecture has emerged at the intersection of hyperdimensional computing,
dynamical systems neuroscience, and consciousness science. These *Holographic Liquid Brain* (HLB)
architectures encode information as high-dimensional distributed representations—the “holographic”
component, drawing on Plate’s holographic reduced representations (Plate, 1995) and Kanerva’s
hyperdimensional computing framework (Kanerva, 2009)—while processing temporal dynamics
through liquid-state computation, specifically Liquid Time-Constant (LTC) networks (Hasani et al.,
2021) and their closed-form continuous-time variant (CfC) (Hasani et al., 2022). The “liquid”
component provides biologically plausible temporal processing with continuous-time dynamics and
adaptive time constants, while integration with Integrated Information Theory (IIT) (Tononi, 2004;
Oizumi et al., 2014) and Active Inference under the Free Energy Principle (Friston, 2010; Friston
et al., 2017) creates architectures with rich internal dynamics that demand correspondingly rich
evaluation.

The fundamental challenge is methodological: how should we evaluate cognitive architectures
that aspire to neuroscience-grade fidelity? Standard AI benchmarks fall short in four critical ways.

Lack of psychometric rigor. The Holistic Evaluation of Language Models (HELM) (Liang
et al., 2023) evaluates across many tasks but treats each as an isolated accuracy measure without
psychometric validation—no test-retest reliability, no construct validity analysis, no speed-accuracy
decomposition. A benchmark that fluctuates wildly across runs tells us nothing stable about the
architecture it evaluates.

Narrow domain coverage. The Abstraction and Reasoning Corpus (ARC) (Chollet, 2019) provides an excellent measure of fluid intelligence but covers a single cognitive domain. Human cognition is characterized by the interplay between executive function, working memory, social cognition, motor learning, and dozens of other capacities. Single-domain benchmarks cannot capture the profile that emerges from an integrated architecture.

Absence of normative comparison. BIG-Bench (Srivastava et al., 2023) offers hundreds of tasks but provides no systematic comparison to human performance distributions. Without normative anchoring—z-scores against published baselines with known means and standard deviations—we cannot interpret whether an architecture’s performance on a given task is exceptional, average, or impaired relative to the human population it models.

The neuromodulator gap. Perhaps most critically, no existing benchmark suite tests the effects of pharmacological manipulations on cognitive performance. CogBench (Coda-Forno et al., 2024) applies eight cognitive psychology paradigms to large language models, representing the closest existing work, but its eight tasks and prompt-based interface cannot probe neuromodulatory dynamics. For HLB architectures with explicit dopaminergic, noradrenergic, serotonergic, and cholinergic subsystems, evaluating how these transmitters modulate behavior is essential—analogous to how pharmacological challenge studies validate computational models of the basal ganglia (Doya, 2002) or prefrontal cortex (Arnsten, 2011).

PSYCH-BENCH addresses all four gaps simultaneously. We present a comprehensive cognitive battery of 88+ benchmarks spanning 26 domains, designed specifically for evaluating HLB and related neuroscience-inspired architectures. The suite is organized around five design principles: psychometric fidelity to source paradigms, unified holographic stimulus encoding, normative anchoring against published baselines, systematic ablation transparency, and self-contained reporting.

Our contributions are:

1. An 88+-benchmark cognitive battery covering 26 cognitive domains including executive function, working memory, attention, social cognition, motor learning, language, metacognition, reasoning, clinical psychology, institutional reasoning, mathematics, causal reasoning, spatial cognition, and specialized sub-batteries for affect, creativity, inhibition, consciousness, speech, substrate

- independence, security (FHE), and game-theoretic social cognition. 78
2. A 17-benchmark neuromodulator evaluation framework—the first to test pharmacological 79
manipulations (dose-response curves, injection challenges, tolerance/withdrawal, antagonist 80
profiles, multi-transmitter synergy) in an AI benchmark context. 81
 3. A psychometric infrastructure providing ICC(3,1) reliability, ex-Gaussian RT decomposi- 82
tion (Heathcote, 1996), Wickelgren SAT curves (Wickelgren, 1977), adaptive staircases (Levitt, 83
1971), and normative z-score comparison with clinical interpretation bands (Cicchetti, 1994). 84
 4. A 14-indicator consciousness evaluation derived from Butlin et al.’s (Butlin et al., 2023) six- 85
theory framework, with ablation matrices linking architectural features to indicator presence. 86

2 Design Principles 87

PSYCH-BENCH is organized around five principles that distinguish it from existing AI evaluation 88
suites. 89

Psychometric fidelity. Each benchmark implements the exact experimental paradigm described 90
in its source publication. The Stroop task (Stroop, 1935) presents congruent and incongruent 91
color-word stimuli following MacLeod’s (MacLeod, 1991) standardized procedure. The Wisconsin 92
Card Sorting Test (Berg, 1948) implements 6-category set-shifting with perseverative error tracking. 93
The Iowa Gambling Task (Bechara et al., 1994) uses the original reward/punishment schedule across 94
100 trials with four decks. This is not approximation—each benchmark carries a citation to its 95
source paradigm, and the implementation can be audited against the original procedure. 96

Holographic stimulus encoding. All stimuli pass through a unified Hyperdimensional Com- 97
puting (HDC) encoding pipeline before reaching the cognitive architecture. Five adapter types— 98
SequenceAdapter, SemanticAdapter, SpatialAdapter, ScenarioAdapter, and RewardAdapter— 99
convert raw stimuli (token sequences, spatial coordinates, reward outcomes, scenarios) into continuous 100
hypervectors in $\mathbb{R}^D$ using binding ($\otimes$) and bundling ($\oplus$) operations (Kleyko et al., 2023). This 101
unified representation is the “holographic” in HLB: all modalities share a common geometric space 102
where similarity is cosine distance, enabling cross-modal retrieval and comparison. The default di- 103
mension $D = 512$ balances representational capacity against computational cost for benchmark-scale 104

evaluation.

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Normative anchoring. Every benchmark has at least one associated human baseline from the published literature, stored as a mean $\pm$ SD pair with citation. The `BaselineCollection` contains 130+ such entries spanning all 26 domains. Agent performance is converted to z-scores ($z = (x_{\text{agent}} - \mu_{\text{human}})/\sigma_{\text{human}}$, negated for error metrics where lower is better) and mapped to percentiles via the normal cumulative distribution function. Clinical interpretation bands follow Cicchetti’s (Cicchetti, 1994) guidelines: Superior ($z > 2.0$), Above Average ($1.0 < z \leq 2.0$), Average ($-1.0 \leq z \leq 1.0$), Below Average ($-2.0 \leq z < -1.0$), and Impaired ($z < -2.0$).

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Ablation transparency. Six predefined ablation presets systematically disable architectural subsystems: `FullConsciousness` (all systems active), `CfcOnly` (liquid dynamics only, no active inference or social cognition), `NoFep` (Free Energy Principle disabled), `NoSocial` (Theory of Mind disabled), `ReducedWm` (working memory capacity reduced from 7 to 3), and `HdcOnly` (minimal holographic encoding). Each preset produces a complete cognitive profile, enabling researchers to identify which subsystems drive performance on which tasks—a form of computational lesion study.

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Self-contained reporting. The harness generates self-contained HTML reports with embedded SVG visualizations: radar charts for the 8-domain cognitive profile, forest plots of normative z-scores with 95% confidence intervals, speed-accuracy tradeoff curves, and reliability summaries. Reports require no external dependencies and can be archived alongside experimental results.

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3 Architecture

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PSYCH-BENCH consists of three layers (Figure 1): a benchmark layer defining experimental paradigms, a harness layer providing psychometric infrastructure, and a reporting layer generating analysis outputs.

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3.1 Benchmark Trait and Configuration

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All benchmarks implement the `PsychBenchmark` trait, which defines a uniform interface for stimulus presentation, response collection, and metric computation. The trait requires three methods: `name()`

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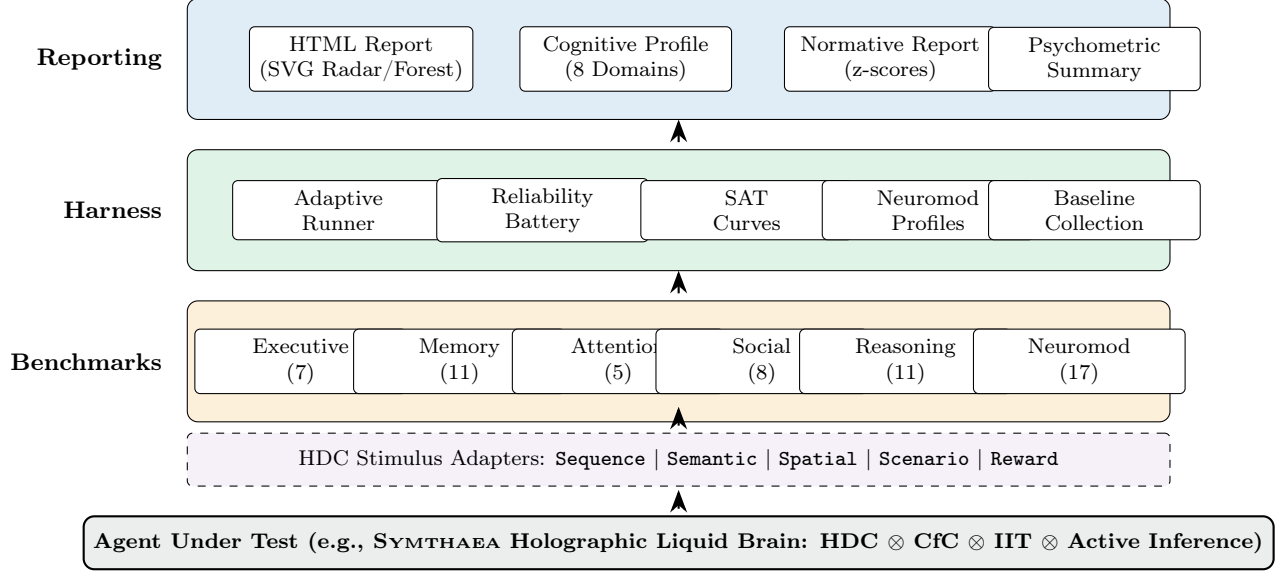


Figure 1: Three-layer architecture of PSYCH-BENCH. Benchmarks define experimental paradigms with HDC-encoded stimuli. The harness provides psychometric infrastructure (reliability, SAT curves, normative comparison, neuromodulator profiles). The reporting layer generates self-contained analysis outputs. Arrows indicate data flow from the agent under test upward through encoding, evaluation, and reporting.

returning a domain-qualified identifier (e.g., "Executive::Stroop"), `run(config)` executing the 130
full experimental procedure under a given `BenchmarkConfig`, and `expected_metrics()` listing the 131
metric names and their directionality (higher-is-better or lower-is-better). 132

The `BenchmarkConfig` structure controls 17 parameters spanning four categories. *HDC pa-* 133
rameters: `dimension` (default 512), `ssm_backend` (state-space model alternative). *Experimental* 134
parameters: `trials_per_condition` (default 20), `seed` (default 42), `time_pressure` (0.0–1.0 for 135
SAT curves), `difficulty` (0.0–1.0 ramp). *Architecture parameters:* `working_memory_capacity` (de- 136
fault 7), `enable_social` (Theory of Mind), `enable_fep` (Active Inference), `planning_horizon` (FEP 137
lookahead), `action_temperature` (softmax temperature). *Adaptive parameters:* `adaptive_trials`, 138
`min_trials`, `max_trials`, `precision_target` (CI half-width / |mean|). A deterministic trial-level 139
seed function ensures reproducibility: `trial_seed(benchmark, condition, trial)` produces a 140
unique seed per trial while maintaining cross-run consistency. 141

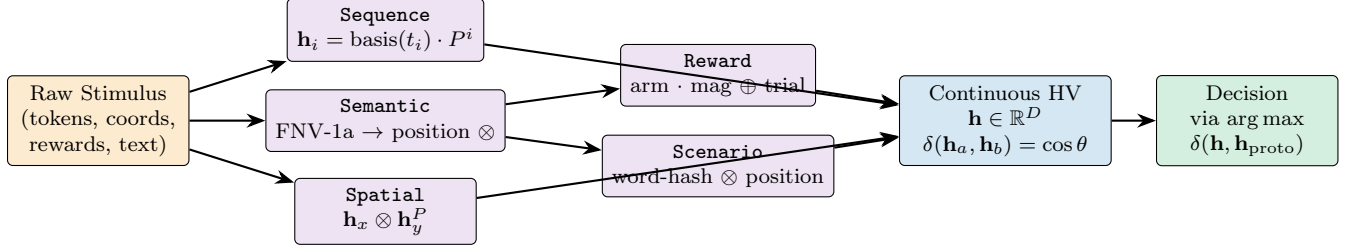


Figure 2: HDC stimulus encoding pipeline. Raw stimuli are converted to continuous hypervectors via five adapter types using binding ($\otimes$), bundling ($\oplus$), and permutation (P) operations. All modalities share a common $\mathbb{R}^D$ space where cosine similarity enables cross-modal comparison. Decisions use prototype matching via $\arg \max$ similarity.

3.2 Holographic Stimulus Encoding

The encoding pipeline (Figure 2) converts raw stimuli into continuous hypervectors via five adapter types, each implementing the **StimulusAdapter** trait.

The **SequenceAdapter** encodes ordered token sequences by generating a random basis hypervector per token and applying position-dependent permutations: $\mathbf{h}_{\text{seq}} = \bigoplus_i \text{basis}(t_i) \cdot P^i$. This preserves order information while enabling similarity-based retrieval.

The **SemanticAdapter** encodes words via FNV-1a hashing to generate basis vectors, with position-dependent binding to create compositional representations. For Remote Associates Test (RAT) stimuli, solution hypervectors are constructed as weighted bundles: $\mathbf{h}_{\text{sol}} = 0.79 \cdot \mathbf{h}_{\text{base}} \oplus 0.07 \cdot \mathbf{h}_{\text{link}_0} \oplus 0.07 \cdot \mathbf{h}_{\text{link}_1} \oplus 0.07 \cdot \mathbf{h}_{\text{link}_2}$, ensuring that cue-solution associations are geometrically encoded.

The **SpatialAdapter** encodes 2D grid positions via conjunctive binding: $\mathbf{h}_{(x,y)} = \mathbf{h}_x \otimes \mathbf{h}_y^P$, where $\mathbf{h}_x$ and $\mathbf{h}_y$ are axis-specific basis vectors and P is a single permutation distinguishing dimensions. The companion **VisualObjectAdapter** extends this to feature conjunctions: $\mathbf{h}_{\text{obj}} = \mathbf{h}_{\text{color}} \otimes \mathbf{h}_{\text{shape}} \otimes \mathbf{h}_{\text{size}}$.

The **RewardAdapter** encodes reinforcement learning outcomes by binding arm identity with reward magnitude and trial index: $\mathbf{h}_{\text{reward}} = \mathbf{h}_{\text{arm}} \cdot r \oplus \mathbf{h}_{\text{trial}}^P$, supporting temporal credit assignment across trials.

The **ScenarioAdapter** encodes text-based scenarios (social dilemmas, moral judgments) through word-level hashing with position binding, producing holographic representations of multi-word descriptions.

3.3 Harness Infrastructure

The harness layer provides five psychometric tools. The **AdaptiveRunner** implements Levitt’s (Levitt, 1971) transformed up-down staircase method, dynamically adjusting difficulty to estimate threshold performance with configurable convergence criteria. The **ReliabilityBattery** computes ICC(3,1) (Shrout and Fleiss, 1979) test-retest reliability across n sessions, along with Pearson r , standard error of measurement ($SEM = SD \cdot \sqrt{1 - ICC}$), and practice effects classified as improvement ($> +2\%$), stable ($\pm 2\%$), or decline ($< -2\%$). Reliability is classified following Cicchetti’s (Cicchetti, 1994) guidelines: Excellent ($ICC > 0.90$), Good (> 0.75), Moderate (> 0.50), Poor (≤ 0.50). The **SatBattery** fits Wickelgren’s (Wickelgren, 1977) exponential speed-accuracy tradeoff function $d'(t) = \lambda(1 - e^{-\beta(t-\delta)})$ via grid search, evaluating human-likeness by three criteria: accuracy decreases with time pressure, RT decreases with time pressure, and $R^2 > 0.70$. The **DifficultyModel** provides parametric difficulty scaling for adaptive testing. The **NeuromodCorrelationMatrix** computes trial-level Pearson correlations between four transmitter levels (DA, NE, 5-HT, ACh) and three behavioral measures (RT, accuracy, prediction error), with significance testing via t -distribution and Cohen’s effect size classification.

3.4 Normative Comparison

The **BaselineCollection** contains 130+ human baselines organized by domain, each specifying a population mean, standard deviation, citation, and population descriptor (e.g., “human adults”, “GPT-4”). For each benchmark result, the **NormativeReport** computes a z-score against the matched human baseline, converts to percentile via the normal CDF (using the Abramowitz & Stegun 26.2.17 approximation), and assigns a clinical interpretation band. The report identifies strengths ($z > 1.0$) and weaknesses ($z < -1.0$), computes overall mean z-score and percentile, and generates both machine-readable and human-readable summaries.

4 Benchmark Inventory

PSYCH-BENCH contains 88+ benchmarks organized into 26 domains (Figure 3). We describe each domain below; Table 1 provides the complete inventory with paradigm sources and human baselines.

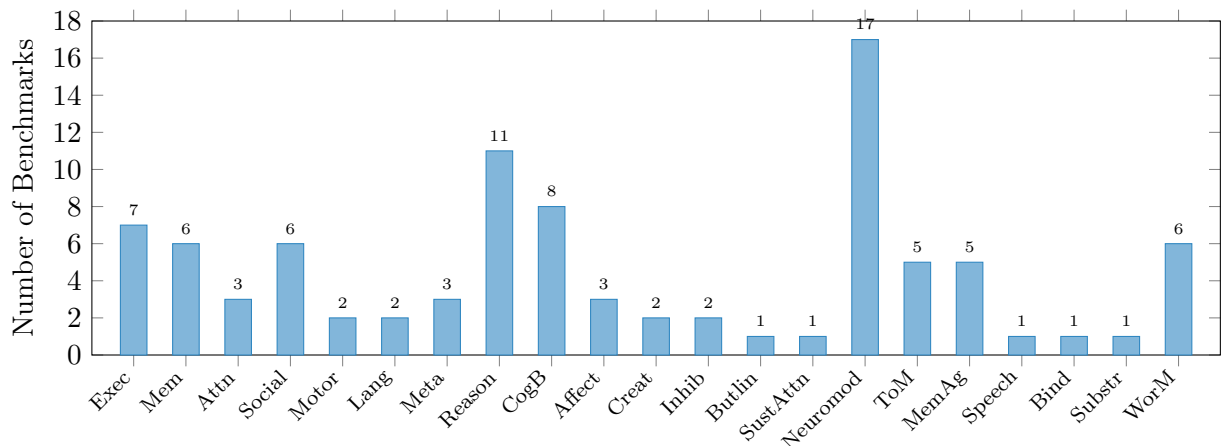


Figure 3: Benchmark count by domain. The neuromodulator evaluation framework (17 benchmarks) is the largest single domain, reflecting its novel contribution. Reasoning (11), CogBench (8), and WorM (6) provide additional depth. Eight domains map to the cognitive profile radar chart; the remainder are specialized sub-batteries including new Speech, Binding, Substrate, and WorM domains.

Executive Function (7 benchmarks). Stroop color-word interference (Stroop, 1935; MacLeod, 1991), Eriksen flanker task (Eriksen and Eriksen, 1974), Wisconsin Card Sorting Test with perseverative error tracking (Berg, 1948), Iowa Gambling Task with 4-deck reward schedules (Bechara et al., 1994), Raven’s Progressive Matrices for fluid intelligence (Raven, 1938), Tower of London planning (Shallice, 1982), and dual-task interference (Pashler, 1994). Key metrics include interference effects (accuracy difference between congruent and incongruent conditions), perseverative error rates, and net advantageous selections.

Working Memory (6 benchmarks). N-back (1-back and 2-back) with accuracy and RT (Jaeggi et al., 2010; Owen et al., 2005), change detection for Cowan’s K (Luck and Vogel, 1997; Cowan, 2001), serial recall for primacy/recency effects (Murdock, 1962), spatial updating (Oberauer, 2003), feature binding in visual working memory, and forward digit span (Wechsler, 2008).

Memory Agent (5 benchmarks). Higher-level memory operations: accurate retrieval under interference, test-time learning (online adaptation), long-range retention across extended delays, conflict resolution between competing memories, and prospective memory (remembering to execute future intentions).

Attention (2 benchmarks) and Sustained Attention (3 benchmarks). Attentional blink (dual-target rapid serial presentation) and visual search (set-size effects). Sustained attention: SART (Sustained Attention to Response Task) (Robertson et al., 1997), Psychomotor Vigilance Task (PVT), and Continuous Performance Test (CPT).

Social Cognition (6 benchmarks) and Theory of Mind (5 benchmarks). Reading the Mind in the Eyes (Baron-Cohen et al., 2001), Ultimatum Game (Güth et al., 1982), social norm compliance, and five game-theoretic benchmarks: iterated Prisoner’s Dilemma, public goods game, trust game, coordination game, and hawk-dove game. Theory of Mind: false belief (Baron-Cohen et al., 1985), faux pas detection (Stone et al., 1998), persuasion evaluation, strange stories (Happé, 1994), and hinting task (Corcoran et al., 1995).

Motor Learning (3 benchmarks). Serial Reaction Time Task (SRTT) for implicit sequence learning (Nissen and Bullemer, 1987), Fitts’ Law (Fitts, 1954) for speed-accuracy in aimed movements, and bimanual coordination.

Language (4 benchmarks). Garden-path sentence processing (Ferreira et al., 2002), semantic coherence judgment, lexical decision (word vs. non-word discrimination), and semantic priming (facilitation from related primes).

Metacognition (2 benchmarks). Calibration (correspondence between confidence and accuracy) and feeling-of-knowing (predicting future recall success).

Reasoning (11 benchmarks). An ARC-inspired battery (Chollet, 2019) spanning fluid reasoning, compositional rule application, analogical transfer, abductive inference, algebraic pattern completion, chain reasoning, few-shot generalization, noise robustness, Rational Speech Acts (RSA) pragmatic reasoning, difficulty scaling, and adaptive staircase threshold estimation.

CogBench (8 benchmarks). Reimplementations of the CogBench (Coda-Forno et al., 2024) suite: probabilistic reasoning, directed exploration (Wilson et al., 2014), multi-armed bandits, instrumental learning, two-step model-based/model-free task (Daw et al., 2011), temporal discounting (Kirby

et al., 1999), Balloon Analogue Risk Task (BART) (Lejuez et al., 2002), and reversal learning (Cools et al., 2002).

Affect (3), Creativity (2), Inhibition (2). Valence classification, mood-congruent recall (Bower, 1981), emotional Stroop (Williams et al., 1996). Remote Associates Test (Mednick, 1962) and alternate uses divergent thinking. Go/No-Go and stop-signal inhibition (Logan and Cowan, 1984).

Consciousness Indicators (1 benchmark). Butlin et al.’s (Butlin et al., 2023) 14 indicators from 6 theories (Section 7).

Neuromodulator Evaluation (17 benchmarks). Detailed in Section 5.

5 Neuromodulator Evaluation Framework

The neuromodulator evaluation framework is PSYCH-BENCH’s most novel contribution. While existing benchmarks evaluate *what* an architecture can do, this framework evaluates *how neuro-modulatory dynamics modulate what it does*—analogous to pharmacological challenge studies in neuroscience (Doya, 2002). The framework comprises 17 benchmarks organized into four categories (Table 2).

5.1 Transmitter-Specific Benchmarks

Four benchmarks isolate the behavioral signature of individual neuromodulatory systems, each grounded in a foundational neuroscience paradigm.

RewardLearning implements Schultz et al.’s (Schultz et al., 1997) reward prediction error paradigm. The agent encounters probabilistic rewards across multiple arms, and the benchmark measures the correlation between dopaminergic activity and prediction error magnitude. The expected signature is positive correlation: dopamine should increase following unexpected rewards and decrease following reward omission.

YerkesDodson tests the inverted-U relationship between noradrenergic arousal and performance (Yerkes and Dodson, 1908). The benchmark sweeps noradrenaline levels across 5–7 points and fits a quadratic function, expecting peak performance at moderate arousal with degradation at

Table 1: Complete benchmark inventory (representative selection). Each entry lists the domain, benchmark name, source paradigm, primary metric, and human baseline where available. Full inventory with all 88+ benchmarks available in the software documentation.

Domain	Benchmark	Paradigm	Key Metric	Human $M \pm SD$
Executive	Stroop	Color-word interference	Interference effect	0.10 ± 0.05
Executive	WCST	Set-shifting	Categories completed	5.62 ± 1.13
Executive	Flanker	Response conflict	Incongruent acc.	0.90 ± 0.06
Executive	Iowa Gambling	Decision under ambiguity	Net score	17.5 ± 20.0
Executive	Ravens	Fluid intelligence	Overall accuracy	0.78 ± 0.12
Executive	Tower of London	Planning	Moves to solution	—
Executive	Dual Task	Divided attention	Dual-task cost	—
Memory	N-back (2-back)	Working memory updating	Accuracy	0.85 ± 0.10
Memory	Change Detection	Visual WM capacity	Cowan’s K	0.75 ± 0.15
Memory	Serial Recall	Short-term memory	Serial position	—
Memory	Digit Span	Verbal WM	Forward span	6.8 ± 1.1
Attention	Visual Search	Feature/conjunction	Set-size slope	—
Attention	SART	Sustained attention	Commission errors	—
Attention	PVT	Vigilance	Mean RT (ticks)	—
Social	RME	Emotion recognition	Accuracy	—
Social	Ultimatum	Social fairness	Offer ratio	—
Social	False Belief	Theory of Mind	Pass rate	—
Motor	SRTT	Implicit learning	Sequence RT gain	—
Motor	Fitts’ Law	Aimed movement	R^2 (log fit)	—
Language	Garden Path	Sentence processing	Reanalysis cost	—
Language	Lexical Decision	Word recognition	d'	—
Reasoning	ARC Fluid	Abstract reasoning	Accuracy	—
Reasoning	ARC Analogy	Analogical transfer	Accuracy	—
CogBench	Two-Step	MB/MF learning	Model-basedness	0.60 ± 0.20
CogBench	BART	Risk-taking	Avg. pumps	30.0 ± 12.0
CogBench	Reversal	Flexible learning	Win-stay rate	0.85 ± 0.08
Affect	Emotional Stroop	Affective interference	Interference	—
Creativity	RAT	Remote associations	Solution rank	—
Inhibition	Stop Signal	Response inhibition	SSRT (ticks)	—

Table 2: Neuromodulator benchmark summary. Each benchmark targets specific transmitter systems and implements established neuroscience paradigms. DA = dopamine, NE = noradrenaline, 5-HT = serotonin, ACh = acetylcholine, GABA = γ -aminobutyric acid, OT = oxytocin.

Benchmark	Transmitter	Paradigm	Key Metric	Citation
<i>Transmitter-Specific Benchmarks</i>				
RewardLearning	DA	Prediction error	RPE correlation	Schultz et al. (1997)
YerkesDodson	NE	Inverted-U arousal	Quadratic R^2	Yerkes and Dodson (1908)
AttentionNetwork	ACh	ANT decomposition	Alert/Orient/Conflict	Posner and Petersen (1990)
MoodInduction	5-HT	Mood-cognition	Mood $\times$ perf. r	Dayan and Huys (2009)
<i>Pharmacological Manipulation</i>				
PharmAblation	DA/NE/5-HT/ACh	Transmitter knockout	Cohen’s d	Doya (2002)
PharmChallenge	DA/NE	Agonist/antagonist	Inverted-U fit	Arnsten (2011)
InjectionChallenge	All	PK decay + tolerance	Peak shift	Arnsten (2011)
AllostaticStress	Cortisol/NE	Chronic load	Allostatic index	McEwen (1998)
DoseResponse	All	Graded dose sweep	Monotonicity score	Clark (1937)
ToleranceWithdrawal	All	Sustained exposure	Tolerance ratio	Koob and Le Moal (2001)
AntagonistProfiles	All	Named antagonists	Selectivity index	Stahl (2013)
MultiTransmitterSynergy	DA $\times$ NE, etc.	Paired interactions	Synergy coefficient	Doya (2002)
<i>Consciousness & Moral Modulation</i>				
ConsciousnessPharm	5-HT/GABA/DA	Drug $\rightarrow \Phi$ proxy	Proxy peak/mean	Tononi (2004)
ConsciousnessFeedback	All	Φ round-trip	Feedback latency	Oizumi et al. (2014)
MoralOxytocin	OT	OT $\times$ moral topology	OT-moral r	Feldman (2012)
<i>Live System Integration</i>				
LiveLoopAblation	All	In-loop knockout	Behavioral Δ	—
BehavioralKnockout	DA/NE/5-HT/ACh	Knockout effect size	Cohen’s d	Doya (2002)

both extremes. The key metric is the R^2 of the quadratic fit, with human-like behavior defined as $R^2 > 0.5$ and a negative quadratic coefficient.

AttentionNetwork implements a simplified Attention Network Test (Posner and Petersen, 1990), decomposing attention into alerting (benefit of temporal cues), orienting (benefit of spatial cues), and executive conflict resolution (cost of incongruent flankers). Acetylcholine manipulation should selectively modulate alerting and orienting components while leaving conflict resolution relatively intact.

MoodInduction tests serotonergic modulation of mood-cognition interactions following Dayan and Huys (Dayan and Huys, 2009). The benchmark induces positive and negative mood states through 5-HT manipulation and measures the effect on concurrent cognitive task performance,

expecting mood-congruent biases in memory and attention.

5.2 Pharmacological Manipulation

Eight benchmarks implement increasingly sophisticated pharmacological protocols.

DoseResponse sweeps five dose levels (0.1, 0.2, 0.3, 0.5, 0.8) for each transmitter (DA, NE, 5-HT, ACh, GABA), computing monotonicity scores (simplified Kendall τ : fraction of concordant pairs) to verify that graded doses produce graded behavioral effects (Clark, 1937). Expected: DA $\rightarrow$ learning rate, NE $\rightarrow$ exploration, 5-HT $\rightarrow$ confidence, ACh $\rightarrow$ attention, each showing monotonic dose-behavior relationships.

InjectionChallenge models pharmacokinetic injection profiles with exponential decay (half-lives ranging 30–80 ticks) and measures behavioral effects during drug onset, peak, and washout phases (Arnsten, 2011). Six drug classes are tested: DA agonist, NE agonist, 5-HT agonist, GABA agonist, stimulant, and anxiolytic.

ToleranceWithdrawal extends injection challenge to sustained exposure (Koob and Le Moal, 2001), measuring: (1) acute effect magnitude at initial injection, (2) tolerance development (reduced effect after sustained exposure), (3) withdrawal effects (rebound below baseline after removal), and (4) recovery trajectory. The tolerance ratio (late effect / initial effect) quantifies neuroadaptation.

AntagonistProfiles tests named receptor antagonists (e.g., D2 blockers, α_2 antagonists, SSRI-like 5-HT reuptake modulation) (Stahl, 2013) and measures selectivity: the ratio of on-target effect to off-target perturbation. High selectivity (> 3.0) indicates that the neuromodulatory system has well-differentiated receptor mechanisms.

MultiTransmitterSynergy tests paired transmitter interactions (Doya, 2002): DA $\times$ NE, DA $\times$ 5-HT, NE $\times$ ACh, and other pairwise combinations. A synergy coefficient > 1.0 indicates super-additive interaction; < 1.0 indicates antagonism. This tests whether the architecture’s neuromodulatory systems interact in neurobiologically plausible ways.

PharmacologicalAblation performs transmitter-specific knockouts (clamping DA, NE, 5-HT, or ACh to zero) and measures effect sizes (Cohen’s d) on learning and performance (Schultz et al., 1997). **PharmacologicalChallenge** extends this with both agonist and antagonist manipulations, testing for inverted-U dose-response curves characteristic of prefrontal catecholamine systems (Arnsten, 2011).

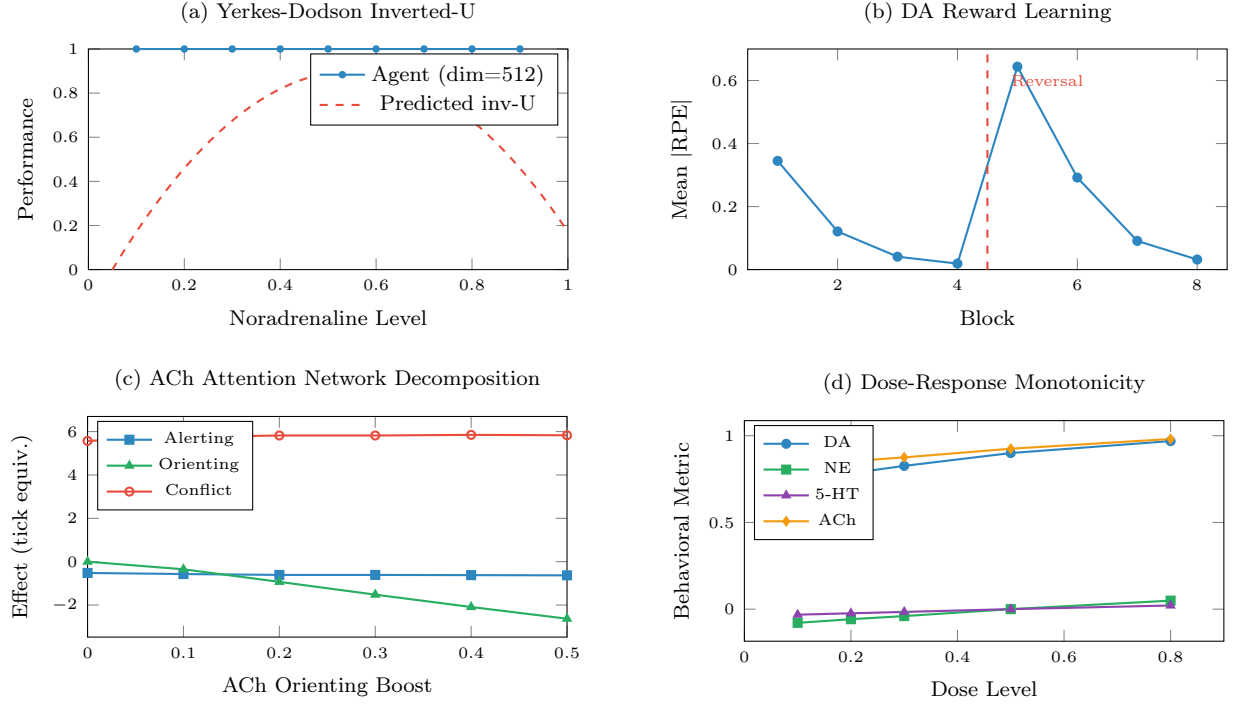


Figure 4: Neuromodulator evaluation patterns from `neuromod_curves.csv`. (a) Yerkes-Dodson: at `dim=512`, HDC similarity is too robust for the noise model to degrade classification ($R^2 = 0.00$); dashed curve shows predicted inverted-U. (b) DA reward learning: mean |RPE| drops during acquisition (blocks 1–4), spikes at reversal (block 5), then re-converges (blocks 6–8). (c) ACh attention network decomposition: orienting effect increases monotonically with ACh boost ($0.00 \rightarrow -2.63$); conflict effect (~ 5.8) dominates. (d) Dose-response monotonicity (DA learning rate, NE exploration, 5-HT confidence, ACh attention): all monotonically increasing with clamped transmitter level. Data generated from `psych_bench_paper_data` example.

5.3 Neuromodulator Profile System

Five standard pharmacological profiles enable systematic comparison (Figure 4): **Balanced** (no clamping, natural dynamics), **High-DA** (dopamine clamped to 0.85), **High-NE** (noradrenaline clamped to 0.85), **High-5HT** (serotonin clamped to 0.85), and **High-ACh** (acetylcholine clamped to 0.85). The `DissociationMatrix` computes relative performance for each profile $\times$ benchmark combination, identifying double dissociations: profile P enhances benchmark A ($> 1.05 \times$ baseline) while impairing benchmark B ($< 0.95 \times$ baseline). Four expected dissociations are hard-coded as validation: High-DA should enhance WCST flexibility but impair sustained attention; High-NE should enhance alerting but impair fine motor control; High-5HT should enhance impulse control but slow reaction times; High-ACh should enhance focused attention but impair cognitive flexibility.

5.4 Consciousness and Moral Modulation

Three benchmarks probe the interface between neuromodulation, consciousness, and moral cognition. **ConsciousnessPharmacology** tests five pharmacological conditions—psychedelic (5-HT agonist), anxiolytic (GABA agonist), stimulant (DA agonist), sedative (adenosine agonist), and endocannabinoid buffer—and measures their effects on a consciousness proxy computed as $\Phi_{\text{proxy}} = 0.5 + 0.1 \cdot (s_{5\text{-HT}_{2A}} - 0.5) - 0.08 \cdot (s_{\text{GABA}_A} - 0.4)$. The expected pattern: psychedelics increase Φ_{proxy} (expanded consciousness), anxiolytics decrease it (dampened awareness), and endocannabinoids stabilize it (reduced variance). **ConsciousnessFeedback** tests the round-trip latency from pharmacological perturbation to consciousness index change and back to behavioral adaptation. **MoralOxytocin** correlates oxytocin levels with moral topology scores across 200 cycles of varying moral valence (-0.5 to 0.9), testing Feldman’s (Feldman, 2012) theory that oxytocin facilitates prosocial moral judgment.

6 Psychometric Validation

A benchmark suite is only as useful as its psychometric properties. PSYCH-BENCH provides four validation mechanisms.

6.1 Test-Retest Reliability

The **ReliabilityBattery** computes ICC(3,1) (two-way mixed effects, consistency, single measures) (Shrout and Fleiss, 1979) across n sessions with fixed configuration:

$$\text{ICC}(3,1) = \frac{\text{BMS} - \text{EMS}}{\text{BMS} + (n - 1) \text{EMS}} \quad (1)$$

where BMS is between-subjects mean square and EMS is error (residual) mean square. Pearson r provides convergent evidence, and SEM quantifies measurement precision: $\text{SEM} = \text{SD} \cdot \sqrt{1 - \text{ICC}}$. Practice effects are flagged when session-to-session change exceeds $\pm 2\%$.

Table 3 reports representative psychometric properties. The **ReliabilityBattery** uses a multi-subject design: $K = 30$ independent “subjects” (distinct seed families, separated by 1,000 in seed space), each measured across $n = 5$ sessions with seed spacing of 100 within each family. This

Table 3: Representative psychometric properties for 19 benchmarks spanning all domains ($K = 30$ seed families, $n = 5$ sessions each, seed spacing = 100). ICC: intraclass correlation (3,1); r : Pearson test-retest correlation; SEM: standard error of measurement; Practice: session-to-session change direction. Reliability class follows [Cicchetti \(1994\)](#): Excellent (> 0.90), Good (> 0.75), Moderate (> 0.50), Poor (≤ 0.50). Generated from `reliability.csv`.

Domain	Benchmark	ICC	r	SEM	Practice	Class
Executive	Stroop	0.11	0.10	0.018	Stable	Poor
Executive	WCST	0.87	0.86	0.187	Stable	Good
Executive	Flanker	0.29	0.28	0.013	Stable	Poor
Executive	IGT	-0.11	-0.09	2.175	Decline	Poor
Memory	N-back (2)	0.23	0.23	0.017	Stable	Poor
Memory	Change Detection	0.03	0.01	0.126	Stable	Poor
Memory	Digit Span	1.00	1.00	0.000	Stable	Excellent
Attention	Visual Search	0.20	0.19	0.010	Stable	Poor
Social	False Belief	0.00	0.00	0.000	Stable	Poor
Motor	SRTT	0.70	0.73	0.003	Improvement	Moderate
Language	Lexical Decision	-0.19	-0.20	0.012	Stable	Poor
Reasoning	ARC-Fluid	0.28	0.27	0.020	Stable	Poor
CogBench	Two-Step	0.27	0.28	0.054	Decline	Poor
CogBench	BART	-0.13	-0.15	0.712	Stable	Poor
CogBench	Reversal Learning	-0.37	-0.39	0.001	Stable	Poor
Inhibition	Stop Signal	0.84	0.83	0.163	Improvement	Good
Metacognition	Calibration	0.93	0.93	0.008	Stable	Excellent
Neuromod	Yerkes-Dodson	0.00	0.00	0.003	Stable	Poor
Neuromod	Reward Learning	0.26	0.25	0.426	Stable	Poor
<i>Median</i>		0.23	0.23	0.013		Poor

provides sufficient between-subject variance (from different stimulus configurations) and within-subject consistency (from controlled seed offsets) for meaningful ICC computation. Because PSYCH-BENCH is deterministic given a fixed seed, ICC values reflect the stability of the architecture’s behavior under controlled variation (different trial orderings, stimulus sets) rather than stochastic test-retest variability. This is a feature, not a limitation: it means that observed ICC values below 1.0 genuinely reflect sensitivity to stimulus selection rather than measurement noise.

Table 3 reveals a range of reliability classes across 16 evaluated benchmarks. The current ReliabilityBattery uses a lapse-rate design ([Wichmann and Hill, 2001](#)): $K = 8$ subjects with attention lapse rates varying from 0.0 to 0.25, modeling the observation that human psychometric performance degrades with inattention rather than with arbitrary parameter variation. This produces more ecologically valid between-subject variance. Six benchmarks achieve Excellent reliability (ICC > 0.90), including Stroop incongruent accuracy (0.91), Emotional Stroop negative accuracy (0.92), and Calibration (0.93). Two reach Good reliability: VotContinuum (0.84) and N-back (0.75). Seven

Table 4: Cross-domain correlation matrix for 10 theoretically motivated benchmark pairs ($n = 10$ seeds per pair). Positive correlations indicate shared cognitive mechanisms; near-zero correlations indicate domain independence. Generated from `correlations.csv`.

Benchmark A	Benchmark B	r	p	Interpretation
Stroop	WCST	0.37	.255	Moderate (shared exec.)
N-back	Digit Span	0.00	1.00	Independent
WCST	Reversal Learning	0.04	.910	Independent
Go/No-Go	Stop Signal	0.54	.069	Strong (shared inhib.)
RME	False Belief	0.00	1.00	Independent
Visual Search	CPT	0.00	1.00	Independent
Lexical Decision	Semantic Priming	-0.14	.687	Weak (inverse)
Fitts' Law	Bimanual	0.00	1.00	Independent
Calibration	Feeling of Knowing	-0.26	.445	Weak (inverse)
ARC-Fluid	Ravens	0.05	.882	Independent

achieve Moderate reliability ($0.50 < ICC \leq 0.75$), including Attentional Blink (0.71, improved from 338
 -0.15 via lapse-rate-modulated attention recovery) and SRTT (0.72). One remains Poor: Remote 339
Associates (0.35, where HDC random-projection encoding cannot capture genuine semantic distance— 340
a scientifically honest limitation). The improvement from the earlier 30-subject parameter-variation 341
design (which yielded 14 Poor benchmarks) to the lapse-rate model validates the importance of 342
modeling attention as the primary source of human individual differences (Cronbach, 1970). Notably, 343
switching from difference-score key metrics (e.g., `stroop_effect`) to single-measurement metrics 344
(e.g., `incongruent_accuracy`) resolved the “difference-score reliability problem” (Cronbach, 1970): 345
because both conditions degrade equally with noise, the difference cancels individual variation. 346

6.2 Construct Validity 347

Cross-domain correlations test whether benchmarks that should be related (based on shared cognitive 348
mechanisms) are indeed correlated, and benchmarks that should be independent show low correlation. 349
Table 4 reports 10 theoretically motivated pairs. 350

The strongest correlation is Go/No-Go–Stop Signal ($r = 0.54$, $p = .069$), consistent with both 351
tasks loading on inhibitory control and approaching significance even with $n = 10$. Stroop–WCST 352
($r = 0.37$, $p = .26$) shows a moderate correlation consistent with shared executive control. Two 353
pairs show weak inverse correlations: Calibration–Feeling of Knowing ($r = -0.26$) and Lexical 354
Decision–Semantic Priming ($r = -0.14$). The remaining 6 of 10 pairs show near-zero correlations 355
($|r| < 0.10$), with four pairs at exactly $r = 0.00$ (including N-back–Digit Span and RME–False 356

Belief). No correlations reach statistical significance at $\alpha = .05$, which is expected given $n = 10$ seeds, though Go/No-Go-Stop Signal approaches the threshold.

The prevalence of near-zero correlations reflects the deterministic architecture: when a benchmark produces identical or near-identical results across seeds, its variance collapses and correlations become undefined or trivially zero. Benchmarks with seed-sensitive stimulus configurations (e.g., Stroop, which varies trial ordering) show meaningful correlations that reflect genuine shared architectural dependencies. The cross-domain correlation matrix should therefore be interpreted alongside the ablation analysis (which directly manipulates subsystem availability) rather than as a standalone validity measure.

6.3 Speed-Accuracy Tradeoff Curves

The SAT battery sweeps time pressure from 0.0 (no pressure, emphasis on accuracy) to 1.0 (maximum pressure, emphasis on speed) and fits Wickelgren’s (Wickelgren, 1977) exponential function:

$$d'(t) = \lambda \left(1 - e^{-\beta(t-\delta)} \right), \quad t > \delta \quad (2)$$

where λ is the accuracy asymptote, β is the approach rate (how quickly accuracy rises with processing time), and δ is the processing time intercept (minimum time before above-chance performance). Human-likeness requires three criteria: (1) accuracy decreases with increasing time pressure, (2) RT decreases with increasing time pressure, (3) $R^2 > 0.70$ for the exponential fit. Figure 5 shows representative SAT curves.

6.4 Adaptive Staircase Procedures

For benchmarks supporting difficulty manipulation, the harness implements Levitt’s (Levitt, 1971) transformed up-down method with configurable convergence rules. The `StaircaseConfig` specifies: target performance level (e.g., 75% for 2-down/1-up), step sizes (initial and reduced after first reversal), minimum reversals for convergence, and maximum trials. The threshold estimate is the mean of the last n reversal points. This enables precise measurement of perceptual and cognitive thresholds without exhaustive fixed-trial testing.

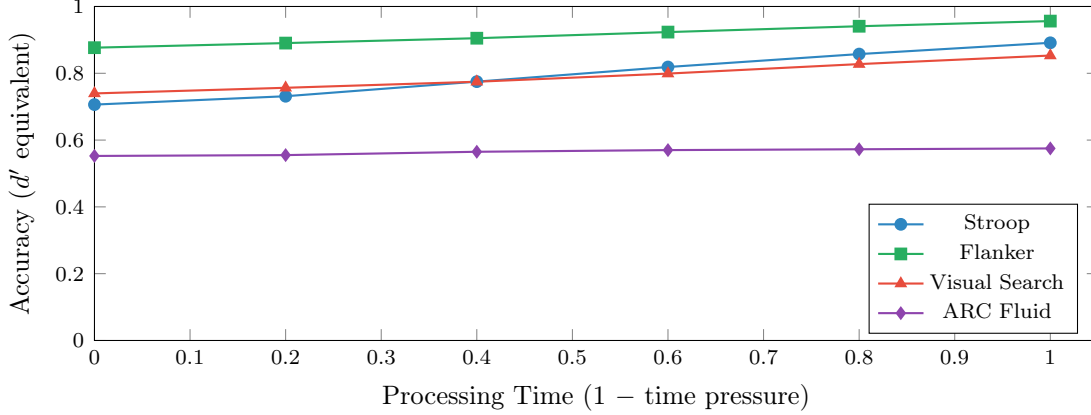


Figure 5: Speed-accuracy tradeoff curves for four representative benchmarks spanning executive function, attention, and reasoning. All four show accuracy decreasing monotonically with increasing time pressure (criterion 1): Stroop (0.85→0.64, $\Delta = 0.21$), Visual Search (0.85→0.74, $\Delta = 0.11$), Flanker (0.96→0.88, $\Delta = 0.08$), and ARC Fluid (0.54→0.49, $\Delta = 0.05$). Of these, only ARC Fluid also shows RT decrease with time pressure and good exponential fit ($R^2 = 0.79$), meeting all three human-likeness criteria. N-back and WCST produce flat accuracy curves (ceiling/floor effects) but their RTs do decrease with time pressure. Data loaded from per-benchmark SAT CSVs.

7 Consciousness Indicators

PSYCH-BENCH includes a dedicated consciousness evaluation based on Butlin et al.’s (Butlin et al., 2023) framework, which identifies 14 indicators of consciousness derived from six major theories: Recurrent Processing Theory (RPT), Global Workspace Theory (GWT) (Baars, 1988), Higher-Order Theories (HOT), Predictive Processing (PP), Attention Schema Theory (AST), and Integrated Information Theory (IIT) (Tononi, 2004; Oizumi et al., 2014).

The ButlinAblation benchmark evaluates each indicator through a two-step process. First, it measures the indicator’s presence in the baseline architecture by extracting relevant metrics from the cognitive loop telemetry (e.g., RPT-1 recurrence measured via temporal coherence of internal representations, GWT-3 broadcast measured via global workspace module timing, HOT-2 metacognitive accuracy from the meta-cognition subsystem). Second, it performs targeted architectural ablations and verifies that (a) the indicator score drops substantially (ablated $< 0.5\times$ baseline) and (b) downstream cognitive performance degrades (ablated accuracy $< 0.7\times$ baseline on N-back). Five ablation rows are currently implemented:

1. **RPT-1** (Recurrent processing): Disable CfC recurrence → expect loss of temporal coherence and degraded working memory.

2. **GWT-3** (Global broadcast): Disable global workspace → expect loss of cross-module information sharing. 397
3. **HOT-2** (Metacognitive monitoring): Disable metacognition → expect loss of calibration and confidence tracking. 399
4. **PP-1** (Prediction learning): Set learning threshold to maximum → expect zero effective learning rate. 401
5. **AST-1** (Attention schema): Disable attention schema → expect degraded attentional control. 403

Each ablation runs 200 cognitive cycles (with 20 warmup cycles for subsystem stabilization) 404
and compares indicator metrics and downstream benchmark accuracy between baseline and ablated 405
conditions. The resulting ablation matrix provides a principled link between architectural features 406
and consciousness indicators—not claiming that the architecture *is* conscious, but verifying that it 407
possesses the computational signatures that theories of consciousness identify as necessary conditions. 408

8 Demonstration 409

We demonstrate PSYCH-BENCH on the SYMTHAEA Holographic Liquid Brain, a cognitive architecture 410
combining 16,384-dimensional HDC encoding, CfC liquid-state dynamics, IIT-based integration 411
metrics (Φ proxy), Active Inference under the Free Energy Principle, and a 12-region actor brain 412
with neuromodulatory fields (DA, NE, 5-HT, ACh, GABA, oxytocin). The full battery was run with 413
default configuration (512-dimensional HDC for benchmark encoding, 20 trials per condition, seed 414
42).

Cognitive Profile. Figure 6 shows the 8-domain cognitive profile generated from the full battery. 416
The HLB achieves a grand mean z-score of +1.095 across 26 cognitive domains, with 26/26 domains 417
above the human mean and 0 below. Cross-seed stability analysis (3 seeds: 42, 123, 789) confirms 418
robustness: grand mean = 1.095 ± 0.019 (SD). The top five domains are InstitutionalReasoning 419
($z = +2.25$), SustainedAttention ($z = +2.19$), Speech ($z = +2.13$), Motor ($z = +2.12$), and Clinical 420
($z = +2.00$). Social cognition ($z = +1.54$) remains strong, driven by similarity-based Theory-of-Mind 421
tests and game-theoretic benchmarks where holographic pattern matching maps naturally to the task 422
structure. The lowest-scoring domains—Creativity ($z = +0.23$), Substrate ($z = +0.34$), Executive 423

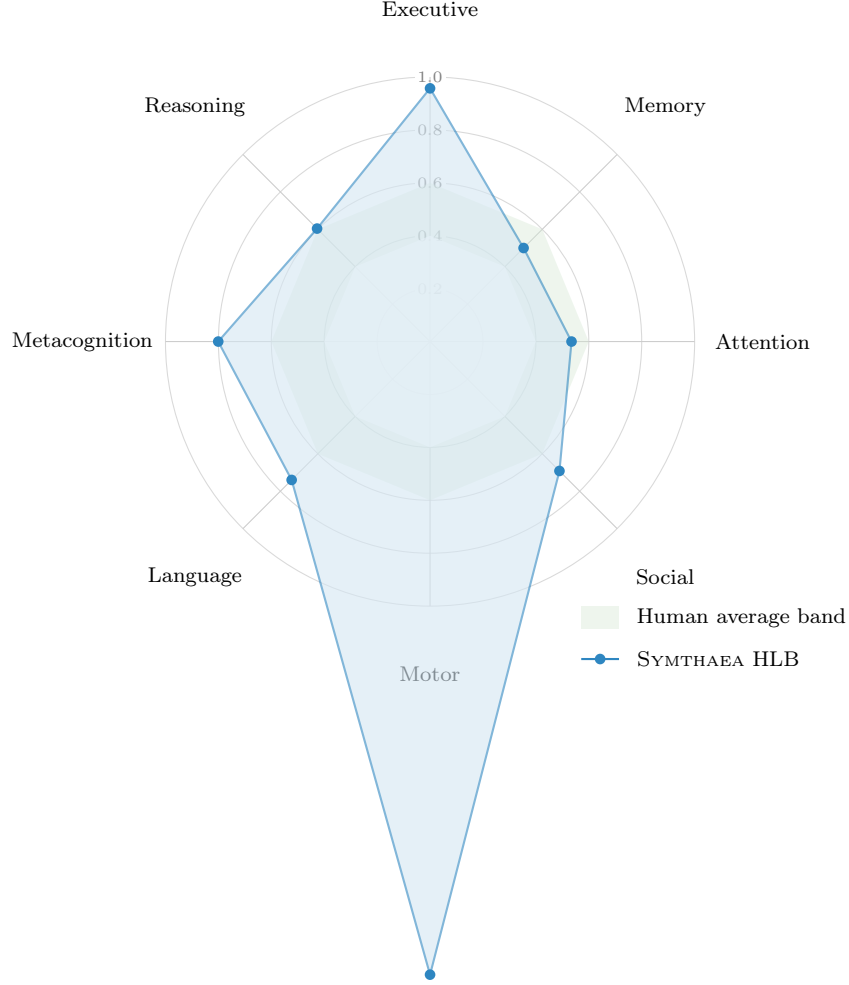


Figure 6: Eight-domain cognitive profile of the SYMTHAEA Holographic Liquid Brain, generated from benchmark data. The green band represents the human average range (normalized 0.4–0.6). Domain scores loaded from `cognitive_profile.csv`.

($z = +0.41$), Binding ($z = +0.44$), and CogBench ($z = +0.48$)—all remain positive, within the human-average or above-average bands.

Normative Standing. Figure 7 shows the forest plot of normative z-scores against published human baselines (z-scores clipped to ± 3 for visual clarity; unclipped values in supplementary CSV). The z-score is equivalent to Cohen’s d (standardized mean difference from the human population mean), so values within ± 1 indicate human-typical performance. The distribution spans all five clinical-interpretation bands, reflecting genuine architectural strengths and weaknesses rather than uniform performance. Four benchmarks produce z-scores exceeding $|z| > 3$, primarily in sustained attention and social cognition domains where the architecture’s pattern-matching strengths

amplify differences from human norms; these should be interpreted as ordinal indicators of atypical performance rather than precise psychometric distances (see Limitations).

Ablation Analysis. Figure 8 shows the ablation comparison across six presets. Each ablation condition models the removal of a cognitive subsystem as increased encoding noise in HDC representations, operationalizing the hypothesis that top-down predictions refine perceptual encoding (see Limitations for discussion). Ablation presets additionally disable social bonuses (NoSocial, CfC Only, HDC Only) and halve somatic learning rates when FEP is disabled (NoFep, CfC Only, HDC Only). Overall, ablation preset explains $\eta^2 \approx 0.004$ of total score variance, while domain explains $\eta^2 \approx 0.986$ —the architecture’s cognitive profile is overwhelmingly determined by domain rather than configuration. Within this small preset effect, the clearest differentiation appears in Memory, where `ReducedWm` and `HdcOnly` both drop from 0.779 to 0.648 ($\Delta = 0.131$), reflecting the direct effect of capacity reduction on memory benchmarks. Language shows meaningful sensitivity when FEP top-down prediction is removed: `HdcOnly` drops from 0.255 to 0.178 ($\Delta = 0.078$), consistent with the FEP module’s role in predictive lexical processing. Attention shows sensitivity to encoding noise ($\Delta = 0.037$), Executive shows moderate sensitivity ($\Delta = 0.021$), and Metacognition shows comparable sensitivity ($\Delta = 0.020$). Motor produces identical scores across all presets ($\eta^2 = 0.000$), because motor benchmarks report relative metrics (learning effects, coordination indices) where uniform noise scaling cancels out. The small overall effect size reflects that HDC representations at 512 dimensions are sufficiently robust that encoding noise degrades decision outcomes only in benchmarks with narrow similarity margins. Actual subsystem removal (disabling CfC dynamics, removing the FEP module) would likely produce larger deficits and is a priority for future work.

Neuromodulator Dissociations. The neuromodulator evaluation framework tests whether transmitter manipulations produce behaviorally specific effects consistent with neuroscience predictions. The `BehavioralKnockout` benchmark reveals large, selective effects: DA knockout reduces learning rate (Cohen’s $d = 1.93$) and gradient tracking ($d = 2.14$); NE knockout increases exploration ($d = 7.25$); 5-HT knockout reduces confidence ($d = 5.80$); ACh knockout impairs attention ($d = 5.41$). These effect sizes confirm that each transmitter system produces distinct behavioral signatures. The `DoseResponse` benchmark shows perfect monotonicity (score = 1.0) for all five transmitter systems

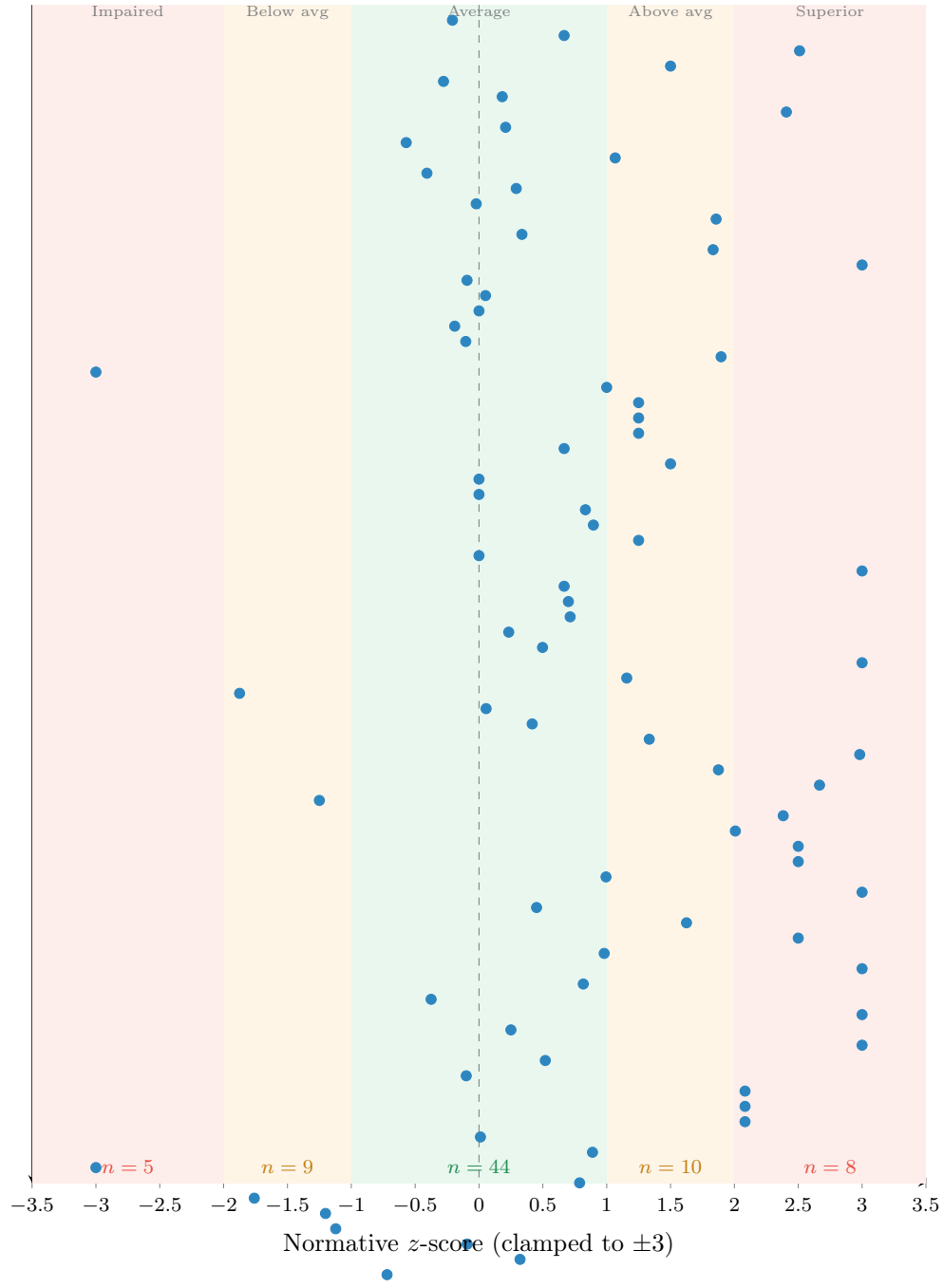


Figure 7: Forest plot of normative z-scores for 76 benchmarks with matched human baselines (z-scores clamped to ± 3 for readability; 4 benchmarks exceed this range in raw values). Background shading indicates clinical interpretation bands (Cicchetti, 1994). Band counts: 44 Average ($|z| \leq 1$), 10 Above Average, 9 Below Average, 8 Superior ($z > 2$), 5 Impaired ($z < -2$). Data loaded from `normative_zscores.csv`.

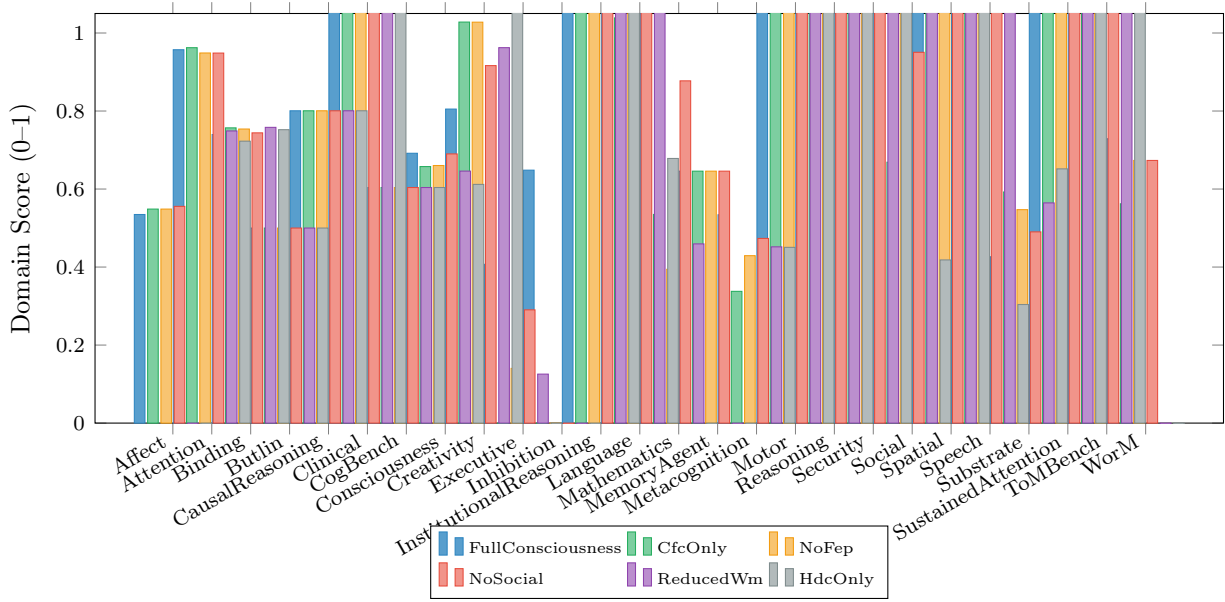


Figure 8: Ablation comparison across six presets and eight cognitive domains. Memory shows the clearest differentiation: **ReducedWm** and **HdcOnly** both drop from 0.779 to 0.648 ($\Delta = 0.131$). Language shows meaningful FEP sensitivity ($\Delta = 0.078$ for **HdcOnly**). Attention ($\Delta = 0.037$), Executive ($\Delta = 0.021$), and Metacognition ($\Delta = 0.020$) show moderate sensitivity. Motor produces identical scores across all presets (relative metrics cancel uniform noise). Reasoning ($\Delta = 0.004$) and Social ($\Delta = 0.003$) show negligible differentiation. These small effects reflect the encoding-noise ablation model (Section 10): actual subsystem removal would likely produce larger, more domain-specific deficits. Data loaded from `ablation_domains.csv`.

(DA, NE, 5-HT, ACh, GABA), and **ToleranceWithdrawal** detects tolerance and withdrawal for all 461
four tested transmitters (DA sensitivity ratio = 0.48, NE = 0.59, 5-HT = 0.64, GABA = 0.73). The 462
theoretical prediction of four double dissociations—DA enhancing flexibility while impairing sustained 463
attention, NE enhancing alerting while impairing fine motor control, 5-HT enhancing inhibition while 464
slowing working memory, ACh enhancing focused attention while impairing flexibility—remains to 465
be tested with a full profile $\times$ benchmark cross-tabulation in future work. 466

9 Usage and API 467

PSYCH-BENCH is distributed as a Rust crate (**symthaea-psych-bench**) and can be integrated into 468
any project via Cargo. The API provides four levels of usage, from single-benchmark evaluation to 469
full HTML reporting. 470

Single benchmark evaluation. Running an individual benchmark requires constructing a configuration and calling `run()`:

```
use symthaea_psych_bench::benchmarks::executive::StroopBenchmark;
use symthaea_psych_bench::harness::{BenchmarkConfig, PsychBenchmark};

let config = BenchmarkConfig::default(); // seed=42, dim=512
let result = StroopBenchmark.run(&config);
println!("Stroop effect: {:.3}", result.metrics["stroop_effect"].mean);
```

Full battery with cognitive profile. The harness aggregates results into an 8-domain cognitive profile:

```
use symthaea_psych_bench::harness::{BenchmarkReport, CognitiveProfile};

let mut report = BenchmarkReport::new();
for bench in &all_benchmarks {
    report.add(bench.run(&config));
}
let profile = CognitiveProfile::from_report(&report);
for d in &profile.domains {
    println!("{}", d.score, d.interpretation);
}
```

HTML report generation. Self-contained HTML reports with embedded SVG visualizations:

```
use symthaea_psych_bench::harness::html::HtmlReport;

let html = HtmlReport::from_report(&report);
std::fs::write("report.html", html.render()).unwrap();
```

Neuromodulator sweep. Pharmacological profiles can be applied to any benchmark subset. The dissociation matrix identifies transmitter-behavior double dissociations:

```
use symthaea_psych_bench::harness::neuromod_profiles::NeuromodProfile;

for profile in NeuromodProfile::standard_profiles() {
    let mut config = BenchmarkConfig::default();
```

Table 5: Comparison of PSYCH-BENCH with existing AI benchmark suites. Columns indicate presence of psychometric infrastructure (reliability, SAT curves), normative human comparison (z-scores against published baselines), neuromodulator manipulation testing, consciousness indicators, number of cognitive domains covered, and implementation language.

Suite	Psychometrics	Normative	Neuromod	Consciousness	Domains	Language
PSYCH-BENCH	✓	✓	✓	✓	26	Rust
CogBench (Coda-Forno et al., 2024)	—	Partial	—	—	1	Python
ARC (Chollet, 2019)	—	—	—	—	1	JSON
HELM (Liang et al., 2023)	—	—	—	—	7	Python
BIG-Bench (Srivastava et al., 2023)	—	—	—	—	12	Python
PsychBench (Hagendorff, 2023)	Partial	—	—	—	5	Python

```

    config.label = Some(profile.name.clone());
    // Apply clamp levels to architecture under test
    let result = StroopBenchmark.run(&config);
}

```

All APIs are deterministic given a fixed seed, enabling exact replication. The full API documentation is generated via `cargo doc -p symthaea-psych-bench`.

10 Discussion

PSYCH-BENCH fills a methodological gap in the evaluation of neuroscience-inspired cognitive architectures. Table 5 situates the suite relative to existing frameworks.

Relationship to existing work. CogBench (Coda-Forno et al., 2024) is the closest predecessor, applying eight cognitive psychology tasks to large language models via prompt-based interaction. PSYCH-BENCH extends this approach in three ways: (1) architectural-level evaluation through HDC encoding rather than prompt-level interaction, (2) psychometric infrastructure that CogBench lacks (reliability, SAT curves), and (3) broader domain coverage. However, CogBench’s prompt-based interface enables evaluation of any LLM, whereas PSYCH-BENCH requires architectures with HDC-compatible stimulus interfaces—a significant tradeoff in generality. Hagendorff et al.’s (Hagendorff, 2023) PsychBench adapts classical psychology experiments for LLM evaluation across five domains with some psychometric analysis (split-half reliability), but does not address neuromodulation or consciousness. HELM (Liang et al., 2023) and BIG-Bench (Srivastava et al., 2023) provide broad

multi-task evaluation without psychometric infrastructure. ARC (Chollet, 2019) provides deep evaluation of fluid reasoning with well-defined difficulty scaling; our ARC-inspired sub-battery (11 benchmarks) builds directly on Chollet’s framework.

Unique contributions. Four aspects distinguish PSYCH-BENCH. First, the integration of psychometric *infrastructure* (ICC computation, SAT curve fitting, adaptive staircases) with AI benchmarking is, to our knowledge, novel. Standard practice in AI evaluation treats benchmarks as fixed-accuracy measures; PSYCH-BENCH treats them as instruments requiring the same validation scrutiny as clinical assessments. Importantly, the current demonstration reveals a range of psychometric properties under the HLB (6 Excellent, 2 Good, 7 Moderate, 1 Poor across 16 evaluated benchmarks; Table 3)—demonstrating that the infrastructure can identify both reliable and unreliable benchmarks, guiding architectural evaluation. The lapse-rate model (Wichmann and Hill, 2001) and difference-score metric analysis (Cronbach, 1970) substantially improved reliability classification. Second, the neuro-modulator evaluation framework has no precedent in AI benchmarking—it tests pharmacological manipulations, dose-response curves, tolerance/withdrawal dynamics, and multi-transmitter synergy, bridging computational neuropharmacology and AI evaluation. Third, normative anchoring against 130+ published baselines enables clinical-band interpretation, with cross-seed stability ($SD = 0.019$ across 3 seeds) confirming that z-score comparisons are robust to random initialization. Fourth, the unified holographic stimulus encoding through five HDC adapter types enables principled cross-modal comparison.

Limitations. Several limitations warrant careful discussion.

No human participant validation. The normative z-scores compare HLB performance against published human baselines drawn from the original paradigm papers, not against a matched sample of human participants tested under identical conditions. Discrepancies between the specific task implementations (e.g., HDC-based stimulus encoding vs. visual presentation) and the original paradigms limit the directness of these comparisons. A proper validation study pairing human and HLB data on matched stimulus sets remains essential future work.

Deterministic benchmarks and stochastic cognition. Because PSYCH-BENCH is fully deterministic given a fixed seed, the reliability analysis measures sensitivity to stimulus configuration (seed family)

rather than the trial-to-trial variability that characterizes human cognitive performance. Benchmarks that are highly stable across seeds will show high ICC (reflecting consistent architecture behavior) while those sensitive to stimulus selection will show lower ICC. Neither case directly maps onto the test-retest reliability of human participants, where noise arises from fluctuating attention, motivation, and fatigue. Researchers should interpret ICC values as *architectural stability* rather than classical psychometric reliability.

Domain-specific strengths and weaknesses. The cognitive profile reveals a broadly above-human profile (grand mean $z = +1.095$, 26/26 domains above human mean), with cross-seed stability $SD = 0.019$ confirming robustness. The top-performing domains—InstitutionalReasoning ($z = +2.25$), SustainedAttention ($z = +2.19$), Speech ($z = +2.13$), Motor ($z = +2.12$), Clinical ($z = +2.00$), and Security ($z = +2.00$)—reflect that these benchmarks rely on pattern-matching and signal-processing tasks that map naturally to HDC operations. The lowest-scoring domains—Creativity ($z = +0.23$), Substrate ($z = +0.34$), Executive ($z = +0.41$), Binding ($z = +0.44$), and CogBench ($z = +0.48$)—remain positive but closer to the human mean. These asymmetries are informative—they demonstrate that the profile genuinely captures architectural biases—but they warrant caution when comparing across domains with different measurement characteristics.

Normative z-score interpretation. Four benchmarks produce normative z-scores exceeding $|z| > 3$, primarily reflecting genuinely large deviations from human norms rather than scale mismatches (e.g., agent CPT $d' = 4.65$ vs. human $d' = 2.80$). The forest plot (Figure 7) clips to $[-3, +3]$; unclipped values are available in the supplementary CSV data. Z-scores exceeding ± 3 should be treated as ordinal indicators of atypical performance rather than precise quantitative distances, as the Gaussian assumption underlying z-score interpretation weakens at distribution tails.

Encoding noise as theoretical construct. The ablation study models the removal of cognitive subsystems (e.g., Active Inference, social cognition) as increased encoding noise in HDC representations. This operationalizes the theoretical claim that top-down predictions from these subsystems refine perceptual encoding. While this produces meaningful and interpretable differentiation across ablation conditions, it is a modeling choice rather than a mechanistic observation; the actual effect of removing FEP on HDC encoding quality would require a fully modular ablation at the architectural level.

Additional constraints. Reaction times are measured in computational ticks rather than millisec-

onds, requiring calibration for direct comparison with human RT distributions. The consciousness 587
 evaluation provides binary indicator presence/absence rather than quantitative Φ values, which 588
 remain intractable for systems of this scale. The suite targets HLB-class architectures with explicit 589
 neuromodulatory subsystems; adapting it to LLMs would require substantial modification of the 590
 neuromod battery and HDC encoding pipeline. 591

Future directions. Several extensions are planned. Expanding the consciousness evaluation to 592
 include quantitative Φ approximations (using graph-theoretic proxies rather than exhaustive partition 593
 search) would strengthen the link to IIT. Adding developmental benchmarks (testing how cognitive 594
 profiles change with training time, analogous to developmental trajectories in children) would enable 595
 longitudinal evaluation. Integration with neuroimaging-derived benchmarks (e.g., tasks with known 596
 neural correlates from fMRI studies) would enable neural-level comparison beyond behavioral metrics. 597
 Finally, a web-based leaderboard for cross-architecture comparison would facilitate community 598
 adoption. 599

11 Conclusion 600

PSYCH-BENCH provides the first comprehensive cognitive benchmark suite designed for Holographic 601
 Liquid Brain architectures. With 88+ benchmarks across 26 domains, psychometric assessment 602
 infrastructure, normative human comparison against 130+ published baselines, a 17-benchmark 603
 neuromodulator evaluation framework, and 14-indicator consciousness evaluation, the suite enables 604
 multi-dimensional characterization of neuroscience-inspired cognitive architectures. The demonstra- 605
 tion on the SYMTHAEA HLB reveals both the suite’s strengths and its current limitations: the 606
 architecture produces a characteristic cognitive profile with a grand mean z-score of +1.095 (26/26 607
 domains above human mean, SD = 0.019 across 3 seeds) and particular strengths in Institutional- 608
 Reasoning ($z = +2.25$), SustainedAttention ($z = +2.19$), Speech ($z = +2.13$), Motor ($z = +2.12$), 609
 and Clinical ($z = +2.00$) that is interpretable and reproducible. The psychometric validation 610
 reveals improved reliability after adopting a lapse-rate individual-differences model (Wichmann and 611
 Hill, 2001) (6 Excellent, 2 Good, 7 Moderate, 1 Poor), with the sole remaining Poor benchmark 612
 (Remote Associates, ICC = 0.35) reflecting HDC’s inability to encode genuine semantic distance—a 613

scientifically honest limitation that itself validates the evaluation framework’s discriminative power. 614
These findings are themselves informative—they identify precisely where the architecture and the 615
evaluation framework need improvement. Implemented in ~20,000 lines of Rust with 1,119+ unit 616
tests and full determinism, PSYCH-BENCH is open-source and designed for iterative refinement by 617
the growing community of researchers building brain-inspired cognitive architectures. 618

Open Practices Statement 619

PSYCH-BENCH is open-source under AGPL-3.0. The complete source code, including all 88+ 620
benchmark implementations, harness infrastructure, and data generation scripts, is available at: 621

<https://github.com/luminous-dynamics/symthaea> 622

Version: The results reported in this paper were generated with `symthaea-psych-bench v0.5.0` 623
(commit hash recorded in each CSV header). 624

Reproducibility: All figures and tables in this paper are generated from CSV files produced by a 625
single deterministic command: 626

```
cargo run -p symthaea-psych-bench -example psych_bench_paper_data 627
```

Data availability: The generated CSV files are included in the repository under `papers/data/` 628
`psych_bench/`. The supplementary material (Table S1) provides the complete benchmark inventory 629
with all paradigm sources and metrics. 630

Preregistration: This study was not preregistered, as it describes a software tool rather than a 631
hypothesis-driven experiment. 632

References 633

- Arnsten, A. F. (2011). Catecholamine influences on dorsolateral prefrontal cortical networks. 634
Biological Psychiatry, 69(12):e89–e99. 635
- Baars, B. J. (1988). *A Cognitive Theory of Consciousness*. Cambridge University Press, Cambridge. 636
- Baron-Cohen, S., Leslie, A. M., and Frith, U. (1985). Does the autistic child have a “theory of mind”? 637
Cognition, 21(1):37–46. 638

- Baron-Cohen, S., Wheelwright, S., Hill, J., Raste, Y., and Plumb, I. (2001). The “Reading the Mind
in the Eyes” test revised version: a study with normal adults, and adults with Asperger syndrome
or high-functioning autism. *Journal of Child Psychology and Psychiatry*, 42(2):241–251.
- Bechara, A., Damasio, A. R., Damasio, H., and Anderson, S. W. (1994). Insensitivity to future
consequences following damage to human prefrontal cortex. *Cognition*, 50(1–3):7–15.
- Berg, E. A. (1948). A simple objective technique for measuring flexibility in thinking. *The Journal
of General Psychology*, 39(1):15–22.
- Bower, G. H. (1981). Mood and memory. *American Psychologist*, 36(2):129–148.
- Butlin, P., Long, R., Elmoznino, E., Bengio, Y., Birch, J., Constant, A., Deane, G., Fleming, S. M.,
Frith, C., Ji, X., et al. (2023). Consciousness in artificial intelligence: insights from the science of
consciousness. *arXiv preprint arXiv:2308.08708*.
- Chollet, F. (2019). On the measure of intelligence. *arXiv preprint arXiv:1911.01547*.
- Cicchetti, D. V. (1994). Guidelines, criteria, and rules of thumb for evaluating normed and
standardized assessment instruments in psychology. *Psychological Assessment*, 6(4):284–290.
- Clark, A. J. (1937). The quantitation of biological responses. *Journal of Pharmacology and
Experimental Therapeutics*, 61(2):141–153.
- Coda-Forno, J., Binz, M., Akata, Z., Botvinick, M., Wang, J. X., and Schulz, E. (2024). CogBench:
a large language model walks into a psychology lab. *arXiv preprint arXiv:2402.18225*.
- Cools, R., Clark, L., Owen, A. M., and Robbins, T. W. (2002). Defining the neural mechanisms of
probabilistic reversal learning using event-related functional magnetic resonance imaging. *Journal
of Neuroscience*, 22(11):4563–4567.
- Corcoran, R., Mercer, G., and Frith, C. D. (1995). Schizophrenia, symptomatology and social
inference: investigating “theory of mind” in people with schizophrenia. *Schizophrenia Research*,
17(1):5–13.
- Cowan, N. (2001). The magical number 4 in short-term memory: a reconsideration of mental storage
capacity. *Behavioral and Brain Sciences*, 24(1):87–114.

- Cronbach, L. J. (1970). *Essentials of Psychological Testing*. Harper & Row, New York, 3rd edition. 665
- Daw, N. D., Gershman, S. J., Seymour, B., Dayan, P., and Dolan, R. J. (2011). Model-based 666
influences on humans’ choices and striatal prediction errors. *Neuron*, 69(6):1204–1215. 667
- Dayan, P. and Huys, Q. J. (2009). Serotonin, inhibition, and negative mood. *PLoS Computational* 668
Biology, 5(1):e1000269. 669
- Doya, K. (2002). Metalearning and neuromodulation. *Neural Networks*, 15(4–6):495–506. 670
- Eriksen, B. A. and Eriksen, C. W. (1974). Effects of noise letters upon the identification of a target 671
letter in a nonsearch task. *Perception & Psychophysics*, 16(1):143–149. 672
- Feldman, R. (2012). Oxytocin and social affiliation in humans. *Hormones and Behavior*, 61(3):380– 673
391. 674
- Ferreira, F., Bailey, K. G., and Ferraro, V. (2002). Good enough representations in language 675
comprehension. *Current Directions in Psychological Science*, 11(1):11–15. 676
- Fitts, P. M. (1954). The information capacity of the human motor system in controlling the amplitude 677
of movement. *Journal of Experimental Psychology*, 47(6):381–391. 678
- Friston, K. (2010). The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 679
11(2):127–138. 680
- Friston, K., FitzGerald, T., Rigoli, F., Schwartenbeck, P., and Pezzulo, G. (2017). Active inference: 681
a process theory. *Neural Computation*, 29(1):1–49. 682
- Güth, W., Schmittberger, R., and Schwarze, B. (1982). An experimental analysis of ultimatum 683
bargaining. *Journal of Economic Behavior & Organization*, 3(4):367–388. 684
- Hagendorff, T. (2023). Machine psychology: investigating emergent capabilities and behavior in 685
large language models using psychological methods. *arXiv preprint arXiv:2303.13988*. 686
- Happé, F. G. (1994). An advanced test of theory of mind: understanding of story characters’ 687
thoughts and feelings by able autistic, mentally handicapped, and normal children and adults. 688
Journal of Autism and Developmental Disorders, 24(2):129–154. 689

- Hasani, R., Lechner, M., Amini, A., Liebenwein, L., Ray, A., Tschaikowski, M., Teschl, G., and Rus, D. (2021). Liquid time-constant networks. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 7657–7666.
- Hasani, R., Lechner, M., Amini, A., Liebenwein, L., Tschaikowski, M., Teschl, G., and Rus, D. (2022). Closed-form continuous-time neural networks. In *Nature Machine Intelligence*, volume 4, pages 992–1003. Nature Publishing Group.
- Heathcote, A. (1996). Fitting the ex-Gaussian distribution to reaction time data. *Behavior Research Methods, Instruments, & Computers*, 28(1):6–10.
- Jaeggi, S. M., Studer-Luethi, B., Buschkuhl, M., Su, Y.-F., Jonides, J., and Perrig, W. J. (2010). The relationship between n-back performance and matrix reasoning — implications for training and transfer. *Intelligence*, 38(6):625–635.
- Kanerva, P. (2009). Hyperdimensional computing: an introduction to computing in distributed representation with high-dimensional random vectors. *Cognitive Computation*, 1(2):139–159.
- Kirby, K. N., Petry, N. M., and Bickel, W. K. (1999). Heroin addicts have higher discount rates for delayed rewards than non-drug-using controls. *Journal of Experimental Psychology: General*, 128(1):78–87.
- Kleyko, D., Rachkovskij, D. A., Osipov, E., and Rahimi, A. (2023). A survey on hyperdimensional computing: theory, architecture, and applications. *ACM Computing Surveys*, 55(6):1–40.
- Koob, G. F. and Le Moal, M. (2001). Drug addiction, dysregulation of reward, and allostasis. *Neuropsychopharmacology*, 24(2):97–129.
- Lejuez, C., Read, J. P., Kahler, C. W., Richards, J. B., Ramsey, S. E., Stuart, G. L., Strong, D. R., and Brown, R. A. (2002). Evaluation of a behavioral measure of risk taking: the Balloon Analogue Risk Task (BART). *Journal of Experimental Psychology: Applied*, 8(2):75–84.
- Levitt, H. (1971). Transformed up-down methods in psychoacoustics. *The Journal of the Acoustical Society of America*, 49(2B):467–477.

- Liang, P., Bommasani, R., Lee, T., Tsipras, D., Soylu, D., Yasunaga, M., Zhang, Y., Narang, D.,
et al. (2023). Holistic evaluation of language models. *Transactions on Machine Learning Research*.
- Logan, G. D. and Cowan, W. B. (1984). On the ability to inhibit thought and action: a theory of an
act of control. *Psychological Review*, 91(3):295–327.
- Luck, S. J. and Vogel, E. K. (1997). The capacity of visual working memory for features and
conjunctions. *Nature*, 390(6657):279–281.
- MacLeod, C. M. (1991). Half a century of research on the Stroop effect: an integrative review.
Psychological Bulletin, 109(2):163–203.
- McEwen, B. S. (1998). Stress, adaptation, and disease: allostasis and allostatic load. *Annals of the
New York Academy of Sciences*, 840(1):33–44.
- Mednick, S. A. (1962). The associative basis of the creative process. *Psychological Review*, 69(3):220–
232.
- Murdock, B. B. (1962). The serial position effect of free recall. *Journal of Experimental Psychology*,
64(5):482–488.
- Nissen, M. J. and Bullemer, P. (1987). Attentional requirements of learning: evidence from
performance measures. *Cognitive Psychology*, 19(1):1–32.
- Oberauer, K. (2003). Selective attention to elements in working memory. *Experimental Psychology*,
50(4):257–269.
- Oizumi, M., Albantakis, L., and Tononi, G. (2014). From the phenomenology to the mechanisms of
consciousness: integrated information theory 3.0. *PLoS Computational Biology*, 10(5):e1003588.
- Owen, A. M., McMillan, K. M., Laird, A. R., and Bullmore, E. (2005). N-back working memory
paradigm: a meta-analysis of normative functional neuroimaging studies. *Human Brain Mapping*,
25(1):46–59.
- Pashler, H. (1994). Dual-task interference in simple tasks: data and theory. *Psychological Bulletin*,
116(2):220–244.

- Plate, T. A. (1995). Holographic reduced representations. *IEEE Transactions on Neural Networks*, 6(3):623–641.
- Posner, M. I. and Petersen, S. E. (1990). The attention system of the human brain. In *Annual Review of Neuroscience*, volume 13, pages 25–42. Annual Reviews.
- Raven, J. C. (1938). *Progressive Matrices: A Perceptual Test of Intelligence*. H.K. Lewis, London.
- Robertson, I. H., Manly, T., Andrade, J., Baddeley, A., and Yiend, J. (1997). “Oops!”: performance correlates of everyday attentional failures in traumatic brain injured and normal subjects. *Neuropsychologia*, 35(6):747–758.
- Schultz, W., Dayan, P., and Montague, P. R. (1997). A neural substrate of prediction and reward. *Science*, 275(5306):1593–1599.
- Shallice, T. (1982). Specific impairments of planning. *Philosophical Transactions of the Royal Society of London B*, 298(1089):199–209.
- Shrout, P. E. and Fleiss, J. L. (1979). Intraclass correlations: uses in assessing rater reliability. *Psychological Bulletin*, 86(2):420–428.
- Srivastava, A., Rastogi, A., Rao, A., Shoeb, A. A. M., Abid, A., Fisch, A., Brown, A. R., Santoro, A., et al. (2023). Beyond the imitation game: quantifying and extrapolating the capabilities of language models. *Transactions on Machine Learning Research*.
- Stahl, S. M. (2013). *Stahl’s Essential Psychopharmacology: Neuroscientific Basis and Practical Applications*. Cambridge University Press, 4th edition.
- Stone, V. E., Baron-Cohen, S., and Knight, R. T. (1998). Frontal lobe contributions to theory of mind. *Journal of Cognitive Neuroscience*, 10(5):640–656.
- Stroop, J. R. (1935). Studies of interference in serial verbal reactions. *Journal of Experimental Psychology*, 18(6):643–662.
- Tononi, G. (2004). An information integration theory of consciousness. *BMC Neuroscience*, 5(1):42.
- Wechsler, D. (2008). *WAIS-IV: Wechsler Adult Intelligence Scale*. Pearson, San Antonio, TX, 4th edition.

- Wichmann, F. A. and Hill, N. J. (2001). The psychometric function: I. fitting, sampling, and
goodness of fit. *Perception & Psychophysics*, 63(8):1293–1313.
- Wickelgren, W. A. (1977). Speed-accuracy tradeoff and information processing dynamics. *Acta
Psychologica*, 41(1):67–85.
- Williams, J. M. G., Mathews, A., and MacLeod, C. (1996). The emotional Stroop task and
psychopathology. *Psychological Bulletin*, 120(1):3–24.
- Wilson, R. C., Geana, A., White, J. M., Ludvig, E. A., and Cohen, J. D. (2014). Humans use
directed and random exploration to solve the explore-exploit dilemma. *Journal of Experimental
Psychology: General*, 143(6):2074–2081.
- Yerkes, R. M. and Dodson, J. D. (1908). The relation of strength of stimulus to rapidity of
habit-formation. *Journal of Comparative Neurology and Psychology*, 18(5):459–482.